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H8SCOOP

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by Henry Fale

BITS & PIECES

SUBSCRIPTION RENEWALS

Just a reminder on how your subscription works. When you receive this newsletter your mailing label contains a number opposite your name. This is the number of your last subscription. The current subscription number is found in the banner such as ISS#1902101081. The first two digits are the current issue, this is issue #19. When you are getting close, send in your \$15 for another year, when you are at your last subscription, the word LAST appears in place of the last issue # on your mailing label. This means that no more issues are mailed out until payment is received. Try not to let this happen, as it could delay your subscription by a month.

I wanted to give fair notice that H8SCOOP will be raising its subscription price soon. Those who want to renew yet at \$15 can do so. If your number on your mailing label is getting close (4 or 5 numbers) to the current issue #, it may be a good idea to renew at the current rate.

ON MX-80 GRAPHICS

A note from Bob Moss from MNET on MX-80 problems with graphics. "It sounds like people are getting their parity bits stripped off... I would suggest that people adapt the H14 method of handshaking for the MX80 only because it permits a greater uniformity amongst users with Heath equipment and allows use of existing drivers and patches (i.e. ctrl P) without significant modification. If you want to use both, this is possible by cabling both and taking one straight off and one off the NOR. Then you can plug into a machine that uses the DSR high=disable without recabling.

For using graphics which uses the 8th bit, you'll have to modify the ASM source code for the driver and reassemble. Modify so the ANI 377 is not stripping the 8th bit.

A TERRIBLE OMISSION

Last month I was forced to cut my editorial short because of lack of space. I wanted to give the subscription information and addresses of BUSS, and REMark for those who do not have this information.

BUSS/ 325 Pennsylvania Ave, SE/ Washington, DC 20003. Pricing is one year which is about 18 issues for \$25. REMark (I hope you all know this one) is \$18/year initial (\$15 renewal) for 10 issues. Address to Heath Users' Group/Hilltop Road/ St. Joseph, MI 49085. Besides these, there are many fine local HUG newsletters that allow the public to subscribe to. I hope to publish a list of some of them in a future issue.

HEATH RUMORS

I have heard rumors (true or false?) that Heath is going to introduce a package called SUPERCAL for about \$200. This is about equivalent to the program helping sell thousands of Radio Shack computers, VISICALC. It's something I have been long waiting for. Also, as reported earlier an 8" 10 Mbyte hard disc with tape capability backup for about \$6000, priced right out of range of most hobbyists. There is expected to be about 25 new software products coming out, many to run under the newly pursued (by Heath) CP/M, which is supposed to be slowly overtaking HDOS. The H8 is to be supported through the middle of 1983 when the new super computer is expected out. There are expected to be more improvements and cards coming for the H8.

Around October, there should be, as reported earlier, the H89 double density card for about \$395, with the H8 card to follow in December. CP/M 2.37 is to be released shortly. I hope it supports those Tandon 400K drives. The H14's price is to drop even more as they try to clear their 2000 inventory to make room for the H25, which is supposed to be a \$1000 kit high quality, high speed dot matrix graphics (same set as H19!) printer. The price is expected to drop to \$395! You may want to pick one up as Epson is struggling to deliver the MX-80 due to its popularity. They can't keep up with demand anymore! There is to be a new serial card for the H89 with 2 serial ports and 1 parallel port to support the MX-80. Look for a 12" green monitor for under \$150, but don't ask me what that is to be used for. The new HDOS 3.0 expected to be released in November of this year, is said to be 0-org as previously reported, and have an MX-80 device driver on it. I guess we'll have to wait and see how much of this comes true-- and on schedule. Oh yes, I also heard DG is supposed to come out with a hard disk drive for the H8, probably under \$2,000.

NPS COBOL?

Captain Steven Tinius sent me an add showing The Software Review, Selling the NPS COBOL for \$69.95. This he can't understand as it is in public domain, unless the \$70 is for the 3-ring binder that comes with it. As reported in a previous H8SCOOP, this is the same COBOL he gives away!

3 DRIVE H17 MOD--HORIZONTAL

Several individuals have written in to me that they are running the H17 3 drive mod putting feet on it and turning it on its side. This way the drives are in the horizontal position instead of vertical which most individuals seem to prefer (and is better for the discs and drives!).

ON SPELLBINDER

I have decided to sell SPELLBINDER at the price of \$295 as mentioned last month. It is the word processor (sold as office management package) that I prepare H8SCOOP on. In my opinion, it beats ANY other word processor out there, costs less, and is MUCH easier to use since it makes use of the special function keys of the H19/H89 (although not necessarily logically). As a dear friend of mine, D. C. Shoemaker puts it "You're completely correct about Spell Binder. Compared to WordStar, this is a truly professional word processor. I'm constantly amused at the folks who go overboard about WordStar (Amen--ed); clearly their experience with the available processors is limited." D. C., by the way, is the one who turned me on to SPELLBINDER--I owe you one Major! He has also been evaluating word processors for the Army.

Anyone wanting to order a package, or receive more information, feel free to contact me.

NEW SOURCE FOR LABELS & DISKS

Bob Sloat writes in that he has had good luck dealing with Disks, Etc./ POB 327/ Center Valley, PA. They have a good price on 5.25" discs and a real good price on labels with tractor feed holes. \$10 and change for a box of 5000 mailing labels.

ON 1.6 and 2.0 INCOMPATIBILITY--H8

D. C. also writes in (See also Request column) "By the way, you might mention that there is an easier way to solve the incompatibilities between HDOS 1.6 and the org-0 modification board. I understand that the patches to the 1.6 discs don't always work, but the following method is infallible and applies to both HDOS 1.6 and CP/M 1.42 and

1.43 disks. The procedure is as follows:

- a) Do a cold-start by pressing RST and 0.
- b) Press MEM.
- c) Set the address to 000.362.
- d) Press OUT.

You're now ready to press GO and boot up the system."

I did! See page 2, May 1981 H8SCOOP! Remember my slogan H8SCOOP--Contact with the future!

NEW HUG CLUB

A HUG is forming in Philadelphia and will be meeting at the Heathkit Electronic Center (Roosevelt Blvd) on the second Wednesday of every month at 7PM. That was sent in by Henry Beechhold.

REPORT ON LINDLEY SYSTEMS MX-80 DRIVER

Brian Hanson wants to pass this information on to you about the Lindley System MX-80 device driver.

For those who want the info, Lindley Systems, 21 Hancock St, Bedford, MA 01730/ Phone (617) 275-6821, price \$20 postpaid. "I was very impressed with the original version 2.1, that I had purchased in early June. Lindley systems has now updated the driver and they gave me the opportunity to update, at a very nominal charge, to their version MXV34.DVD.

"I would like to tell you some of the features which have impressed me. All printer controls are programable with normal text characters. The text can control Double Strike, Condensed and Emphasized print modes. Form length and automatic line skip at page break can be used if desired. Spaces can be added to the left margin if desired. The driver can set the high bit so that programs (like B.H. Basic) that output seven bits to print files, can access the graphic mode in the printer. Either DSR or CTS, RS232 protocol can be used. The driver can access 5 device names (LP0:, LP1:, LP2:, LP3:, and LP4:) through a single I/O port and each has a 16 character initialization sequence that is preprogrammed but can (also) be reprogrammed to any desired sequence, such as 40 characters per line normal print, 66 characters per line normal print or any other configuration desired."

ON MAGNOLIA CP/M

I'm asked about CP/M so often, who has what, and what will handle the Tandon drives I want to add this section. It deals with the Magnolia CP/M, since this is what I mostly get questions about.

The Magnolia Microsystems (of whom I am a dealer) CP/M with their double density card for the H89 has a CP/M that will run only with their board, and only with the H89 to give the standard drive about 180K CP/M and the Tandon about 750K. This card is only for double density H89, and since all their CP/M BIOS use Z80 code, it is not transferable to the H8.

They have a single density package, again using Z80 codes (but A converted H8 CPU could handle this) and using the standard Heath drives. Although they specify the H89, I see no reason why it would not run on a Z80 H8. It will support the Single Density Tandon drives giving about 350K for CP/M.

There are still those out there working on the CP/M BIOS for the Heath CP/M, but I haven't heard any word on the progress of these. One thing to keep in mind, anyone working on this project should know what Magnolia is doing, since they have set the "standard" in non-standard Heath drives. I'd sure hate to see a half dozen different modified CP/M's going around that were not compatible. We have enough compatibility problems, let's leave it at that.

I also hear the Magnolia CP/M will support only 128 directory entries, while Heath's will support 256 (for their new version).

GOOD PRICES ON PRODUCTS

Lum-Buckles, a Zenith factory-authorized distributor, is holding an inventory reduction sale. All prices include shipping to a street address. Orders may be prepaid or COD (COD with 20% deposit, add 3% for COD). The items offered are

Z-89-FA (disk, 48K, serial I/O) for \$2100*

Z-89-GA (48K, serial I/O) for \$1900*

Z-87 dual drive for \$850*

Z-47 dual 8" drive with controller for \$2500

Diablo/Xerox 630 printer for \$2000*

Z-19 terminal for \$706

CP/M, HDOS, or AutoScribe for \$125 each

Scotch or BASF 5" 10 sector discs 10 for \$26, next 10 for \$23

Diablo Hytype II film MS ribbons 12 for \$45, next 12 for \$40

Diablo Titan 10 metal print wheels for \$35

Starred items--order 2 or more, get free dust covers. Quantities limited to stock on hand. Order from Lum-Buckles/ 7 Gateview Ct/ San Francisco, CA 94116. Phone (415) 731-7793.

ONLINE CLEARINGHOUSE FOR MICRO INFO

A new source of information and advice on micros is now online. Called "The Micro Advisor", it is accessible through the CompuServe Information Network (Micronet) to anyone with a terminal or personal computer. It contains features of interest to anyone using or thinking about using microcomputers, including individuals, professionals; executives, small business people, etc.

To find out more about it, play around with it. You can access it by typing "TMA" while in CIS mode. Others interested should contact Battery Lane Information Services/ POB 30214/ Bethesda, MD 20814.

DATA TERMINALS CORRECTION

Data Terminals & Communications/ 590 Division St/ Campbell, CA 95008/ (408) 378-1112 wrote to inform me that the information published in last month's H8SCOOP (page 3) on **Re-Conditioned Keyboard Diablo Printers** was incorrect. The refurbished 300's sell for \$1295 plus tax and shipping, and the DTC 382's sell for \$2,000 plus tax and shipping. The 300 uses a Diablo HyType I print mechanism while the 382 the HyType II print mechanism.

NEW PRICES ON 400/800K TANDON'S

Because of my increased quantities of ordering the Tandon drives, I am now able to reduce my price to you. Single quantity of drive alone is \$460, the second for \$450. Quantity orders can be discounted greater than before, write or call for complete information.

I'm still waiting for the modified CP/M BIOS for these drives. Magnolia has several configurations if you want to buy the whole CP/M over again, but so far nothing for the H8. What happened to all you guys that said you'd have the Heath BIOS modified? I haven't heard a word on it as of yet, and there are lot of individuals waiting.

UPDATE FROM MI-8

MI-8/ 822 E. County Rd. 30/ Ft. Collins, CO. 80525/ (303) 669-4116 has a price reduction on the V-8 Model II speech synthesizer for the H8. It is currently being sold for \$324.95 and includes full software (3 disks). In November, there may be another H8SCOOP exclusive on the PC-12 parallel board with a special 3 port price when purchased with the V8.

A question has come up which deserves some space. I have

been asked, and so has Ted of MI-8, so I'll let Ted tell it. "With the introduction of various voice systems available for most computers as well as the Heath, one most frequently asked question is "Which one is best for the Heath computers?". I would like to say the V-8 Model II is without a doubt. However, the answer depends on the users' application and requirements of his voice system.

"For a low cost limited application, the Digitalker can fulfill a good number of requirements. The Digitalker consists of two ROM's containing 143 words and phrases along with a speech processor. The voiced output is very high quality. The vocabulary is broad enough to meet a great number of applications. A talking clock, volt meter, voice prompted terminal, and a good number of talking educational programs, such as math quizzes and spelling. The Digitalker also requires very little programming.

"For the person requiring more versatility and unlimited vocabulary in English or any language, the V-8 Model II would have to be the logical choice. I am often asked how the V-8 compares to the Type-N-Talk. My response would have to be that there is little comparison. While Type-N-Talk can speak with virtually no programming, it lacks the intelligibility of the V-8. Both use the SC-01 speech synthesizer chip, but in a different way. The Type-N-Talk produces its English speech with the use of a ROM English algorithm. The algorithm is about 90% accurate, which is quite good when dealing with the many rules of the English language. The speech it produces is, however, a direct result of that algorithm and may not be the speech the user would like to hear. On the other hand, the V-8 uses a look-up dictionary of words and speaks them exactly the way the user wants to hear them. Any word can be changed and any number of words can be added. Changing or adding a word in the dictionary is quite simple with the utility "SPEAK" program. SPEAK allows the user to configure a new word to sound the way you like before adding it to the dictionary. An example of this would be the word "coffee" which can be pronounced in any number of ways depending on where you live. Anywhere from upstate New England or Southern California and anywhere in between. The user can also add inflection to his words and sentences to give them the clearest sounding speech available today."

"MI-8 has gone to great lengths to provide software to support the V-8. There is virtually no software provided with the Type-N-Talk. (There is a device driver disk available with demo programs and sounds from the Keyboard Studio, by Tom Jorgenson--ed). Our programs include a 1,000 word dictionary, which is user expandable, and several utilities to support the V-8. There is also the device driver "VB.DVD" which allows the user to add speech to any BASIC program or other language that recognizes device drivers. There are also several BASIC programs for the V-8 user. A talking clock and talking game puzzle. There is more coming in from V-8 owners everyday. To further encourage this, MI-8 is offering a full disk of programs to anyone submitting a program of their own along with written permission to redistribute it.

"I can honestly say there is no other voice system available for the Heath computer that offers the versatility and support of the V-8 Model II. After all, we are entering our second year of voice for the Heath computers. But don't take our word for it, ask around, or better yet, compare yourself. For versatility, support, and voice experience with Heath computers, there is no comparison.

"The V-8 Model II sells for \$249.95 kit or \$349.95 assembled and tested. The Digitalker sells for \$119.95 kit or \$149 assembled and tested." To order, contact MI-8.

NEW STUFF

MORE FROM HOYLE & HOYLE

BAUDOT to DISK receiving and correcting programs for \$15 are available from Hoyle & Hoyle Software/ 716 S. Elam Av/ Greensboro, NC 27403 (NC residents add 4% sales tax). This program runs on the H8, H17 and H19 combo requiring HDOS 2.0 and 16K memory (I guess the H89 will also work?--ed).

Using the Baudot to Disk Receiver, your H8, a Ham rig and the standard interface between the Ham rig and the H8, you will be able to receive and decode 60 or 100 wpm baudot/murray code. The program is also set up to receive 110 and 300 baud ASCII code. The info is stored in a buffer which can be saved on your disc.

Features include the following: Prints on the screen what you are saving, clears the screen instantly by pushing one key, single key changes receive speed at any time, 10 automatically named files, protection for already named files, stores incoming data when buffer is full, easy buffer clear to store new data at any time, clear, constant display of free space in disk and buffer.

A simple, small screen editor is included to correct pictures, with the special ability to strike over characters on the same line by giving the user the option of eliminating the LF on any lines desired. It allows you to read in 20 lines at a time, make corrections on the screen and store the corrected version.

This program package is ideal for receiving and editing pictures or copy (Including Military transmissions and press bulletins) received on Ham operating equipment. A small collection of pictures received and edited using these programs is also included on the disk.

DISK DUMP UTILITY FOR CP/M 2.2

New from Elektrokonstult/ Konnerudgaten 3/ N-3000 Drammen/ NORWAY for \$29.95 including disk and documentation (not source code) is DDUMP, a sector oriented disk dump utility which makes it possible to examine and modify any byte(s) on any sector, addressed by track and sector number, OR addressed by allocation block (or 'group') number. Sector contents are printed and/or displayed - and may be patched - in both HEX and ASCII formats. DDUMP allows you to dump and patch data not otherwise accessible by CP/M, and it may, for example, be used to examine text files for control characters, tabs, etc., to study how CP/M allocates disk storage, examine and repair a damaged disk and recover deleted files and lost data.

DDUMP works with most types of drives and disks used with CP/M, including 5" and 8", hard and soft sectored, single and double density, single and dual sided disks. The program automatically adapts to the actual number of tracks and sectors per track for most types of diskettes without manual intervention.

To order, include \$4 postage and handling. Check or VISA accepted.

H8 BUSS PC BOARD HEAT SINK EXTENSIONS

From MI-8/ 822 E. County Rd. 30/ Ft. Collins, CO 80525/ (303) 669-4116 comes a "HOT" item for the H8, Heat sink extensions for the PC board. It's a well known fact that a fully loaded H8 generates a considerable amount of heat. On some boards the 7805 5 volt regulators operate close to their thermal shut-down limit, and the convection cooling of the H8 can be quite insufficient. Heat is one thing that can damage expensive computer IC's, which was the reason for many H8ers installing fans in their H8.

MI-8 has developed a simpler and quieter solution, the addition of HEAT-SINK EXTENSIONS. They connect to the PC board mounting brackets and can be installed in just a few minutes, without any modification to the H8 enclosure. Tests have shown up to a 65% of heat reduction on some boards. These extensions are made of fast heat

dissipating aluminum and come complete with mounting hardware. Price is \$3.25 for 1 to 3, \$2.99 for 4 to 6, and \$2.49 for 7 or more. Include \$2.50 handling and shipping per order.

INEXPENSIVE 8" CONTROLLER

Here's more info on the 8" controller I mentioned last month. This is available from Livingston Logic Labs/ 2386-323 E. Del Mar Blvd/ Pasadena, CA 91107. The 8" disk controller is very simple, made up of only 14 IC's, the controller chip being the Western Digital WD1771. Total cost of parts should be under \$100 from most suppliers.

In operation, the controller is virtually identical to the H-47, except that it does not support double density. The total disk capacities for the two formats supported by this controller are 8" single side--250K, double side is 500K. There is a version for the H89 and the H8, but since it uses Z80 non maskable interrupt capability, the H8 must have a Z80 board or a Z80 mod to the CPU. In addition to this, you must have at least 32K RAM, 0-org mod, and 1-4 Shugart compatible 8" drives. Presently, CP/M is required and supported, and proof of ownership MUST be supplied before the software can be shipped to operate this system. This is necessary because much of the software, including the BIOS is derived from standard CP/M, and Ray Livingston doesn't cherish the thought of spending time in jail for copyright violations. This may change in the future as software "patches" are being developed to work with CP/M. Hopefully the next few months will produce the HDOS drivers to operate this under HDOS as well.

Your present 5" controller can be used as normally for your 5" drives. When you purchase the controller you receive the controller board (the same size of standard H8 boards for the H8, and slightly shorter than the H89 boards for the H89), complete documentation including schematic, parts list, source code, a replacement monitor ROM which is identical to Heath's except the H-47 boot routines have been deleted and replaced by the routines necessary to boot from any drive connected to this disk controller. The new monitor ROM operates exactly the same way your standard ROM (boot from any device, etc.). In order to keep costs low, you are requested to send your old ROM in after receiving the new one so it can be reused (there is a deposit). A complete set of software is also supplied, which includes: Bootstrap loader, Bios loader, Bios, MOVCPM, CONFIGUR, FORMAT--all come with complete source code except MOVCPM which comes with source code for the modified part.

Pricing as follows. No personal checks only cashiers or money order. For the H89 systems, the basic price is \$75, \$200 assembled, and \$15 ROM deposit. The H8 basic price is \$85, \$215 assembled, and \$25 ROM deposit. This ROM deposit is refunded when you send back the ROM (if you elect to do this). The IC's needed to build the kit are commonly available small scale TTL logic type IC's. Specify the disc you want your software on, 5" or 8". If you want 5" add \$15 as it requires several of them to hold all the software. Projected dates are September 15 for H89 controller, and October 1 for H8 controller. If you are interested and would like to be placed on the waiting list for the order, contact Ray Livingstone of the above mentioned address. Do not send any money until you hear from Ray, as he will inform you when things are ready.

Other products in the works are new serial/parallel I/O board for the H89 which will contain 2 serial ports and 3 parallel ports using the 8255. A high resolution color graphics interface for the H89 is planned for under \$200.

ALSO EPROM BURNING SERVICE

If you need an EPROM programmed, Ray can do it for \$5. Just send him the EPROM, or he'll supply one at a reasonable price. Also send the software in one of the following formats: HODS .ASM or .ABS, CP/M .ASM, .MAC, .HEX, .PRE, .REL, or .COM, or an already

programmed ROM or EPROM. If either .PRE or .REL format is supplied, specify the origin address you desire. If disk is supplied, use either Heath 5" or 8" Single density (single or double sided). Any questions on any of the above, Ray can be reached by phone at (213) 792-4763.

SOFTWARE FROM REICHERT DIGITAL SYSTEMS

Henry Beechhold told me he purchased two programs from Reichert Digital Systems (29 Blazier Road, Warren, NJ 07060), both of which are winners. LOGIC, is a digital-logic simulation program designed to produce a truth table showing the state relationships of all outputs of a digital circuit for all possible input conditions. The program has numerous features, including an editor, and runs under HDOS MBASIC. The cost is \$29.95.

The other is really two programs, XYLOT and BARLOT, both of which are sophisticated plotting programs. This package sells for \$29.95. These can be used whenever data is needed to be plotted on screen or printer.

SOFTWARE TOOLWORKS T-SHIRTS

For \$6.50 you can have the Software Toolworks T-shirt featuring a picture of Leonard, their H89 dreaming of a floppy disk and their moto, "Without software, it's just a paperweight". It's silk screened in black on gold color, 100% cotton. They may be ordered in mens size S, M, L, or XL. These are top quality shirts so they run slightly larger than most. These are available only from Tri-Arts/ 7854 Lankershim Blvd./ North Hollywood, CA 91605, and will be shipped only in the 48 contiguous states. California residents add tax. Price includes shipping--allow 3 to 4 weeks for delivery.

CP/M BUSINESS SOFTWARE FROM VANDATA

For those looking for business software on the H89 format, contact Vandata/ 17544 Midvale Ave. North/ Suite 205/ Seattle, WA 98133/ (206) 542-7611. They have available the complete Osborne business package consisting of General Ledger with Cash Journal (\$295), Accounts Receivable (\$295), Accounts Payable (\$295), and Payroll with Cost Accounting (\$295). Buy all four for \$995. These require CBASIC2 2.05 or higher which is available for \$110, and CP/M 1.4 or higher.

These require an 8080 or Z80 system with minimum 48K RAM, double density 8" or two smaller capacity disk drives. Three diskette drives are the optimum configuration. The format supplied and supported is the 5 1/4" single-density format. A 132 column printer is required for printout, and the H19 or H89 or a similar type terminal with addressable cursor.

This is a much modified, enhanced and debugged version of the original Osborne package. Comes complete with very adequate documentation and user manuals.

They also have available for \$700 dBASE II a relational database, and PEARL III for \$650, a program generator. VISA, MC, and COD orders accepted.

8" DISK SYSTEM FOR H88'S AND H89

Floppy Disk Services/ C.N. 5212/ Princeton, NJ 08540/ (609) 771-0374 has available for H88 and H89 owners a disk drive system using the CDR controller card. It is capable of single and double density at the same time. This is presently for CP/M (I think--ed). They can supply the BIOS to drive the thing if you have CP/M. The complete package including 2 single sided double density drives, cabinet, hardware, manuals--everything needed is \$1600. Complete with single drive in double cabinet is \$1300. Assembled and tested controller only is \$600.

H19/H89 AUTO REPEAT AVAILABLE

From R Peterson/ POB 122/ Livermore CA 94550, for \$60

comes Auto-Rep-89. This allows the H19 or H89 terminal to automatically repeat any key by holding down the key, thus eliminating the need for holding down the key and the repeat key simultaneously.

Any key will repeat at a specific rate, after a 300ms delay to avoid double strikes. No modifications are made to your terminal. It's simply a small module that plugs into the existing connector on the keyboard, and the other end to the cable that used to go to the keyboard (in series). No external power is required, and its use in no way affects the normal operation of the terminal.

APPOINTMENT CALANDER BY SUNFLOWER

Sunflower Software/ 6304 W 76th St/ Prairie Village, KS 66204 has for 39.95 an executive appointment calander package eliminating the need for an appointment book. This package is comparable for some selling as high as \$295. It's called EASY for Executive Appointment Secretary. It is written in ABS so execution is fast. There are three modes of operation, Calander Mode, Full Screen Mode, and Sectioned Mode.

In the Calander mode, the screen displays a full representation of the current month with the prompt at today's date. There are a multitude of options available, with the numeric key pad allowing you to move forward or backward a day at a time. An "H" is displayed on dates for holidays, and a "V" may be placed in days for vacation.

At this point, the blue key takes you to the Full Screen Mode, from which you may enter appointments of messages and edit them with the numeric key pad. Any information on the screen in this mode can be printed. Pressing the "shifted 7" key will cause the times of the day to be superimposed on the screen in 15 minute intervals starting with 8AM. This is the "Sectioned Screen Display Mode", making the appointments more meaningful.

The times displayed are protected fields and may not be deleted. When you delete a line from this mode, only the appointments themselves are affected. This looks like an excellent package for the businessman or busy hobbyist, for a very affordable price.

CROSS REFERENCE FOR REMark

REMREF is available for \$12 at most Heathkit stores or from Keyboard Studio. What is REMREF? It is a data disk for use with Keyboard Studio's Keyword Index Program, KEY-IT. REMREF is a data file containing author, title, keyword information for all articles appearing in REMark issues 1 through 20. With REMREF together with KEY-IT, you can quickly obtain all references in REMark to a specific subject, author, or other keyword. It is designed so that the reference keys not only tell you which REMark to look in, but the starting page number of the article! Also, since REMREF is an ASCII text file, it can be edited to include future issues of REMark. It is available in both CP/M (REMREF-C) or HDOS (REMREF-H). For more information, contact Keyboard Studio/ 125 Aspen/ Birmingham, MI 48009 (313) 645-5365, or the author, Dave Powers/ Box 1154/ Troy, MI 48099 (313) 879-1115. Bob Ellerton of REMark tells me in a near future REMark, the entire REMark index will be printed, FYI.

NEW STATUS PROGRAM

In the beginning, there was the STAT command of ver 1.6 of HDOS. Some useful information, but not much! Then came version 2.0--some new useful information but what happened to all the old information? NEWSTAT is a program that gives the user more complete details of their system status, which includes all the information of 1.6 and 2.0 (except DVD status), in both split octal and decimal. Also, the types of disk errors are broken down for hard and soft errors. It is interesting to note that all errors are not reported by the system as errors in the STAT command--this program reports ALL detected errors! Also included is

the last volume number mounted and track accessed for each drive. This program will, if errors are encountered, ask the user for permission to clear all errors. As an assembly option, the user can elect to have all errors cleared after displaying them or to eliminate entirely the error clearing routines. If the user has any routine that puts the time (in ASCII) in the 'S.TIME' cell, this program will print it in its header. The source code is included. Runs on ver 1.6 or 2.0 H8/H89. For a copy send \$15 or \$12 and a disc to Gregory Marsh/ 5279 Miles Ct/ Woodbridge, VA 22193.

NEW ROOTS BY COMMSOFT

COMMSOFT/ 665 Maybell Av/ Palo Alto, CA 94306/ (415) 493-2184, is releasing ROOTS/M. It is almost identical to ROOTS89 which is currently offered by Heath and is COMMSOFT for HDOS, but will run on CP/M or MP/M and is available on 8" or 5.25" disks. The program supports a number of CRT terminals including the H19 and H89.

ROOTS/M allows you to store, retrieve and display thousands of facts about your family. It uses standard genealogical conventions to print pedigree charts (family trees) and family group sheets. With the program you can prepare a print out of your family trees back to 1AD! Another of the seven modules in the program allows you to compute the relationship between individuals. With the aid of ROOTS/M you can settle the argument about whether an individual is your grand aunt or second cousin once removed.

ROOTS/M comes with an illustrated typeset manual which provides an extensive course in how to trace your family tree. The price for ROOTS/M or ROOTS89 is \$124.95. For more info, contact Howard Nurse of COMMSOFT.

NORTHSTAR BASIC CP/M ON H89?

BaZic is a BASIC interpreter that will allow any H89 equivalent computer to run any NorthStar BASIC program under CP/M, and run it faster and more efficiently. Since it uses Z80 code for optimization, if you want to run it on an H8 you would have to have the Z80 CPU. It is available for \$150 from Micro Mikes Inc/ 3015 Plains Blvd/ Amarillo, TX 79102.

Micro Mike's claims that baZic is 28 to 60% faster than North Star BASIC. It provides several terminal control facilities within the language such as statements to clear the CRT screen and position the cursor at a specified location. The greatest improvement over North Star BASIC is the executable APPEND statement, which enables you to write a software package containing many different programs and APPEND subroutines common to more than one module while the program is running. The advantage of this is each common subroutine need appear on the program disk only once, which results in disk space gained, and program maintenance is much easier. INFOWORLD, April 27, 1981 rated Usefulness as excellent, Documentation as good, ease of use as good, and error handling as excellent.

MONEY MAESTRO HOME BANKING SYSTEM

InnoSys Inc, a leading supplier of software to the banking industry since 1973 is pleased to announce its entry into the personal computer market with its MONEY MAESTRO home banking system. This new package written in FORTH under CP/M is designed to provide financial recordkeeping, tax reporting and budgeting for professionals, clubs, families and very small businesses. Those who recognize the benefits of accurate records but don't require full-blown double-entry accounting systems will find MONEY MAESTRO to be the fast, simple and complete money management package they seek.

Based around the user's checkbook, the system will record all deposits, payments, and charges by category (or group of categories) and payee while simultaneously keeping and accurate check register. These records are supplemented by entries of all financial transactions which don't affect

the account balance (charge cards, cash payments, etc.) so that MONEY MAESTRO can give complete reports on budget status, payment histories, and all income or expenses by category. Its powerful stockpiling feature will allow the preview of bills before payment and will hold partial payments or regularly occurring payments for later payment.

For more information, or to order, contact InnoSys. Price of user's manual alone is \$20, price of complete package is \$200. Specify 8" or 5.25" CP/M disk. The address is 2150 Shattuck Av/ Berkeley, CA 94704, phone (415) 843-8122.

BIBLIOGRAPHY UPDATE

September 1981 Microcomputing had an article written by Kathleen and D.C. Shoemaker (he's got the wife in the act now) titled "Relief for the Hassled Clerk". In the article they describe a timekeeping program whereby you enter timekeeping information by clock and job number. It is written in MBASIC under HDOS. This starts on page 162.

Another on page 92 by James Ingram "Tame That Blackboard Jungle" is a program written under MBASIC and HDOS for an H8/H17/H9, although he states H19 modification is straightforward. The program contained performs three main functions, entry of activities, insertion and deletion of activities, and display of current information. It was written mainly for a teachers use in a class, where a disk record is kept of every day and date of every actual class throughout the school year.

OFFERS

I make all software available an an as-is basis, in other words, no money back policy if not satisfied, mainly because it's too easy to copy discs. You take your chances. At my prices, you really can't go wrong. I try to get software out as fast as possible so you can get the newest stuff fast, and as thus, don't get a chance to carefully screen and test everything. To do so, would delay all software at least 3 to 4 months. I do the best I can and sell without assuming responsibility on the accuracy or runability of a program or disc. I don't think I've let anyone down in the past, and will continue to do my best not to.

The following are past offers made by H8SCOOP, which are and will continue to be made available to those interested readers. Note I have dropped a few items from the list, and as time goes on, more will be dropped so my inventory doesn't go crazy around here.

Tandon Drives W/Software	\$485
Tandon Drive only	\$460
2ND. Drive	\$450
Single Cabinets W/PS	\$100
H89 DD Magnolia board W/CP/M	\$510
H89 CP/M For 400K Tandon +16K add on	\$250
SPELLBINDER Office Management Pkg.	\$295

PROJECT ARTICLES FOR PARALLEL PORT--WRITE FOR CATALOG

9/80 File Dumps & Utilities.	\$10
11/80 Pictures, Lables, Word Processor.	\$15
12/80 Pictures & Utilities.	\$10
1/81 H19 Graphics program disc.	\$10
2/81 Plant, Swarms, Starwars, Utilities.	\$10
3/81 Potpourri Set(2).	\$15
3/81 Adventure Set(2) Pirates, Lost Dutem.	\$15
4/81 THINGS--H19 Graphics, Biorythm, Utility	\$10
5/81 HDOS 2.0 PATCHES	\$10
7/81 MISC I--OREGON TRAIL, ATOM20	\$10
7/81 ODDS & ENDS--Sort, Calculator, Finance	\$10
8/81 HVC; Bigfont (excellent), cake, chair	\$10

If the bootlegging of these software offers does not cease, this column will be discontinued. Once that happens, it will

not be re-instated! When only 20 discs sell to my entire readership, I know something is wrong. This makes it not worth my effort, or my contributors effort.

This months OFFER 9/81 is MIXED BAG OF SOFTWARE for \$10. It contains a modified version of my Selectric device driver to work with 2 or 4MHz computers, settable baud rates, ports, and echo or no echo to console, and uses the H8-4 4 port serial card. The driver was written long ago as one of my project articles, back in the days when I had time to do assembly programming, even though I did not even know what an assembler was months before that. It has come a long way since then. It also contains some programs of interest to those using the 400K drives. EDUMP is a very quick one line change to fix HUG's DUMP which allows dumping of all 160 tracks. PNDUP.ABS is a relatively major modification to HUG'S DUP program which allows copying between any two drives and will copy more than the normal 40 tracks. Both drives must be the same type. It also preforms a byte by byte comparison on verification and will display track, sector an byte of any difference. PNDUP19 is the same, but for the H19 terminal. DATE.ACM is an assembly language program which will code and decode an ASCII date into the HDOS two-byte format.

TECHNICAL FORUM

CUSTOMIZED BOOTABLE HDOS DISKS

This item is printed by permission from CHUG, Capital Heath Users' Group, Inc., POB 341/ Fairfax VA 22030. CHUG is an independent local HUG with USA and Canada addresses membership rates of \$12 and all others (foreign) \$25/year. It is an excellent newsletter to have. This article is taken from August 1981 page 4.

Although most users of the HDOS Operating System are now familiar with the use of PROLOGUE.SYS files, there is a little known method for creating "customized" bootable HDOS disks which uses a similar technique. When HDOS boots up, it checks to see if a file named PROLOGUE.SYS exists on drive SY0:. If the file is found, it is automatically executed when the booting process is completed, thus providing the user with a pseudo turnkey capability. This technique can be carried one step further, however:

- 1) INITIALize and SYSGEN an HDOS disk in the usual manner.
- 2) Mount the SYSGENed disk on a drive other than SY0:
- 3) Remove the Lock flag from the file SYSCMD.SYS using the HDOS DUMP program to modify the DIRECT.SYS file (or use an equivalent method).
- 4) DELETE the file SYSCMD.SYS!
- 5) COPY an ordinary HDOS.ABS file to the SYSGENed disk.
- 6) RENAME the .ABS file to SYSCMD.SYS.
- 7) Install the modified disk in drive SY0: and boot HDOS.

At this point you will have created a SUPER PROLOGUE.SYS! When the disk is booted, the program names SYSCMD.SYS will automatically be executed at the end of the boot process. The big difference, however, is that when the program terminates, it is automatically reexecuted again! This is precisely the way that the normal HEATH supplied SYSCMD.SYS receives control when a user program finishes executing.

It is impossible to perform any operation other than execute the program named SYSCMD.SYS with this configuration. Typing a double Control/Z sequence merely starts the program at the beginning again!

Two obvious uses for this technique are:

- 1) Create a limited operating enviornment for an unsophisticated user who only needs to know how to operate a single program.
- 2) The replacement SYSCMD.SYS can be written as a MENU driven selection program which is used to specify which program from a set of programs on the disk is to receive

control via a LINK SCALL. When the selected program terminates, control will be returned to the MENU "SYSCMD.SYS" and the next program (or function) can be specified. Two programs which illustrate the use of this technique are ZENITH Electronic Typing (AutoScribe) and the newer HUG Dungeons and Dragons game.

It should be noted that all unnecessary files can be deleted from the bootable disk since they cannot be executed from HDOS command level. The only files really required are the new "SYSCMD.SYS", HDOS.SYS, HDOSOVLO.SYS, HDOSOVLI.SYS, SY.DVD (for V2.0) and possibly ERRORMSG.SYS.

GOOD GUYS and BAD GUYS

GOOD GUYS--ALL HEATH OWNERS & USERS ACROSS THE WORLD. This nomination came in from Janet Hoyle of HOYLE & HOYLE SOFTWARE. Janet writes "Back in May when we started a software business our personal banker told us to expect 10% "bad" checks and strongly advised us not to ship to anyone until the check cleared. Well, we ignored that advice and shipped to everyone who ordered, the same day. As we expected, in over 500 orders, we have had no checks returned for insufficient funds or any other reasons. More than just good financial relations, our customers have been good to meet (via mail & phone) and pleasant to deal with. We enjoy the 2 to 3 phone calls a week asking for help or hints. The letters from people with questions or frustrations are still friendly and complimentary, even if they complain that we misspelled scabbard or left out an ending quotation mark, etc...(Yes we always have to have the spelling critics--gives them something to do--ed).

"In our new, soon to be released A Galactic Experience any word misspelled is deliberate because that's the way it's 'posed to be in the far reaches of outer space... Please express our thanks to all the fine folks who use Heath." I must say I agree with Janet, I have had the same experience. The majority of my subscriptions, offers, and drives are shipped by check, before they ever clear, and I have not yet to be burnt. It makes me feel good that in this crazy mixed up world we live in, the Heath group are indeed "GOOD GUYS"--a rare breed of human beings that play it straight (mostly--see editorial) and are excellent people to talk to, meet, and work with. My getting into this field has been the most rewarding experience of my life, a move I have never regretted. Hats off to all you Heath owners and users.

BAD GUYS--BEAL AGENCY/ 4141 Fairmont Ave/ San Diego, CA 92105. Georges Troupin ordered a disk doubler kit for \$10 to convert his discs to double sided discs and hasn't heard a word from them. They were mentioned in May 1981 H8SCOOP under NEW STUFF. This is the first offense against them, so we'll kind of play it cool and see what happens.

BAD GUYS--ADVANCED COMPUTER PRODUCTS. This outfit has been written up twice already, July 1981 and April 1981 H8SCOOP. It appears this column may be neglected, and if that is the case I will discontinue it. Send in your feedback, either you want this column or you don't. I'm not writing this for my health, I have plenty of other things to do, and a lot more to fill H8SCOOP with besides this column. When I started out I choose this column to try to help guide my subscribers to who is worthy of our support, and lot of feedback tells me this column is being neglected. If that's the case, I'll stop printing it. Let me know your views.

Anyway, Gordon Richards ordered some Verbatim discs from them, and has problems. The price was good and he received them in good time, but after using some he discovered some would not mount because they were 16 sector rather than 10. Even the label said 10 sector (wait till you hear my story --ed). After contacting ACP, he was

told since the order was over 30 days old, they could not exchange or help in any way, he should have checked them ALL right away! Now what kind of bull is this? I buy discs in the hundreds, and I check them as I use them. If a company has this reputation, plan to spend lot of time checking an order after receiving it--in this case impractical! Anyway, suffice it to say they offered no satisfaction.

His next call was to Verbatim, who said this should be handled by the store selling them, but in this case they would exchange them. (They are the GOOD GUYS) in this case. There has been much BAD stuff written in about ACP, for your own good, stay away from them. You may have had some good luck, but don't count on it continuing!

Now for my experience. The same thing happened to me using Memorex. My 10 sector hard discs had a 10% error giving me 1 sector soft, even though labeled 10 sector hard. I got these from PACIFIC EXCHANGE (see next GOOD GUYS) and they promptly shipped me my replacements. I also wrote a letter to Memorex and they never even had the decency to respond or acknowledge my letter! If they didn't have such reliable and good discs, I would not be buying from them. Especially after they claim every disc is checked before it leaves the plant--hogwash!

GOOD GUYS--PACIFIC EXCHANGE/ 100 Foothill Blvd/ San Luis Obispo, CA 93401, (1-800-235-4137) or (805)-543-1037 in California. I mention them because this is the only place I now buy discs from, and have not only been totally satisfied, but have not had one complaint against them. They handle various types of disks in all kinds of configurations--your one stop headquarters. They ship COD UPS, after your first order Cashier check, they will ship Company check OK. You call in an order, and it usually arrives within 5 working days. Very impressive. They advertise lowest prices available (which they are) and stock the items (which a lot don't).

There was one time they were out of stock. What did they do? They shipped me double side 10 sector hard sectoring for the same price as standard single side. Now that's what I call a GOOD GUY. They go out of their way to help you. I recommend all disc users to go through them as they won't let you down. Small or large quantities, you get the best prices and the fastest service.

GOOD GUYS--Hoyle & Hoyle Software/ 716 S. Elam Ave/ Greensboro, NC 27403. Tom Kyle writes in to tell me how they go out of their way to help him, and I have heard this from several others. For \$1 to cover postage and handling plus the original disc they provide an updated version of their "A Remarkable Experience" adventure game. The updated version corrects some bugs that existed at the end of the game.

Tom has also purchased their second offering "Steer Kleer", which is a real time absolute, variable speed, graphic, manual dexterity skill game, requiring the player to navigate a ball through an obstacle course. For \$10 one receives a 5" disc containing 2 assembled games, 2 unassembled games, plus the documented source code. They have been more than helpful. Judging by their new offers, they are on the way to the top, and that comes by being a GOOD GUY. You can have all the software and hardware you want, but your sales and life are based on whether you are a good guy or a bad guy. This is what separates the men from the boys.

GOOD GUYS--William Campbell, MD/ 855 Smithbridge Rd/ Glen Mills, PA 19343. Doc Campbell sells an assembly language primer for the beginner that covers some of the basics for HDOS assembly programming. Cost is \$15. He also provides a 3 page addenda to clarify some points in the main papper. I have to agree. Reading some of his articles in REMark, and just my dealings with him nominate him for the GOOD GUYS column.

Tom adds--I have yet to be disappointed with any outside

(Heath) product I have purchased for the H8. The budget doesn't allow me to buy many of them (the story of our lives), but when I do part with some of my rare extra bucks for a product, it is gratifying to find a supplier that provides the extra support for the product and purchaser.

CLASSIFIEDS

FOR SALE--H11A with 32K RAM, 2 Serial I/O and 2 adapt cables. Original cost over \$2000, will sell entire setup for \$1500 or best offer. Also H8 with 3 16K RAM boards for a total of 48K. Sell boards for \$230 each, or best offer. Write for more information. John Hong/ Box 85/ Nome, AK 99762.

FOR SALE--Godbout 12K Static memory board for \$60. Heath H8-5 Serial/Cassette interface with manual and cassette software for only \$50. Heath Assembly Language Programming Course for \$25. Will sell all for \$110. Thomas Bessire/ 7600 Pacific Place, #2015 Bld. Ft. Worth, TX 76112. Phone (817) 451-3912.

FOR SALE--DG 32D RAM Board less memory chips, \$150. Heath 8K Ram board \$50. Omnitec Modem- Acoustic, 300 Baud \$60. Michael Couch/ 1218 18th St/ W. Des Moines, IA 50265. Phone (515) 223-0549.

FOR SALE--Z89 FA factory built H89, brand new, for \$2200. Also 100K Seimens disk drive, steps at about 12ms for \$150. Frank Rinkleff/ 2036 E. Ridgeway Rd/ Waterloo, IA 50702. Phone (319) 234-6566.

FOR SALE--12K Godbout RAM for \$100, 8K Heath Ram for \$60. 2-Heath 8K RAM boards without memory chips or sockets for \$25 each. H8-7 breadboard without breadboard sockets for \$40. Gordon Richards/ POB 7211A/ Orlando, FL 32854. Phone (305) 298-8729.

FOR SALE--Complete Heathkit H8 based 48K system. Includes H19 and H9 terminals, H17 dual disk, H8-5 serial I/O, 16K extra RAM, H8-4 multiport serial I/O, H8-2 parallel I/O, H8-7 breadboard, HTB-0 Mullen extender board, 25 disks with programs, 16 new disks, interface cables, Zenith Word Processing, Much software including Microsoft and Extended BH BASIC, HDOS 2.0, many cassettes, new Diablo 1640 with extra type/fonts and 21 carbon ribbons. All manuals. All in excellent condition. Best offer over \$2000 on H8 system, and best offer over \$2000 on Diablo. Robert Ackerman/ 5870 Country Ln/ Aberdeen, SD 57401. Phone (605) 255-7057.

FOR SALE--1 DG Electronics DG-32D 32K Ram board for H8 fully populated, factory wired \$250. 2 Heath WH8-16 16K Ram boards for H8 factory wired \$155 each or both for \$290. Allan Moser/ Route 1, Box 16/ Samuels, ID 83862. Phone (208) 263-3726.

FOR SALE--Two WH-8-16 16K memory boards, \$200 each. Jack Van Zandt/ 350 Sharon Park Dr., Apt Q-4/ Menlo Park, CA 94025. Phone (415) 859-4718 during day.

WANTED--Z89 computers (used-below \$1500) at a reasonable cost. Contact Allen Bernson/ POB 68/ Boston, MA 02167.

FOR SALE--H8 up to 40K memory, with or without H8-5, H8-2 and PC-12 interface boards. Make offer. Ted Benglen/ 822 E. County Rd. 30/ Ft. Collins, CO 80525. Phone (303) 669-4116.

FOR SALE--NOGDS HA8-2 music synthesizer board for H8, complete with on board 1 watt stereo amplifier, sufficient for two 8" speakers with an output level greater than one would likely use at home. 3 discs of software, schematics, and full documentation also included. \$85 or best offer. D. C. Shoemaker/ 2000A Foxridge/ Blacksburg, VA 24060. Phone (703) 552-5764.

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NOVEMBER issue deadline--October 16

Subscription Rates--\$15/Year (\$21 overseas Airmail)--current issues

Back issues #'s 1, 2, 3, 7, 8, 9, 10, 11 no longer available. Remainder through March 1981 (issue 12) still available at \$1 each while supply lasts. Volume 2 back issues available at \$1.25 each to current issue.

PHONE HOURS: 9AM to 9PM as I am available, except sunset Friday to sunset Saturday. NO COLLECT CALLS.